GitHub + VS Code | New Contributor Guide

Windows & macOS • Repository: github.com/AkashM07/ComputerVisionTask

1. Clone the project (once)

```
git clone https://github.com/AkashM07/ComputerVisionTask.git
cd ComputerVisionTask
code .
```

clone downloads the repository and its history; **cd** enters the folder; **code .** opens that folder in VS Code (or use File → Open Folder). Do not run **git init** after cloning.

2. Start a branch from the latest main

```
git switch main
git pull origin main
git switch -c feature/description
```

Update your local **main**, then create and switch to a new branch for your changes. Use a descriptive branch name, such as **feature/image-processing**.

3. Edit and commit locally

```
git status
git add .
git commit -m "Describe your changes"
```

Edit scripts in VS Code. **status** reviews changes; **add** stages them; **commit** saves a local snapshot. Check that generated datasets and secrets are excluded before staging.

4. Push your branch (first time)

```
git push -u origin feature/description
```

origin is the GitHub remote; **-u** sets the upstream tracking branch. It links your local branch to its matching GitHub branch, so future **git push** and **git pull** can work without specifying the remote and branch. Without an upstream, a first plain push may fail (unless auto-setup is configured).

5. Later pushes and pull requests

```
git push
```

After the first push, **git push** is normally enough. On GitHub, open a **Pull Request** from your feature branch into **main** for review and merging.

6. Refresh main after merging

```
git switch main
git pull origin main
```

Pull teammates' merged changes before starting your next feature branch.

Local vs. GitHub

Each contributor has an independent **local copy**. Editing and committing do **not** update GitHub; **push** uploads commits. Others receive shared changes with **pull** after they are pushed/merged.

SSH authentication fix

If you see **Permission denied (publickey)**, switch the remote to HTTPS:

```
git remote set-url origin https://github.com/AkashM07/ComputerVisionTask.git
```

Check with **git remote -v**, then retry pushing. GitHub may ask you to sign in.